

Babies Statistical Enquiry - Printable Worksheet

AS91946 (1.1) PPDAC relationship investigation

Student worksheet

2026-08-09

Name: _____

Class: _____

Investigative Question

What is the relationship between birth weight and age of the mother?

Use this worksheet to complete a full PPDAC enquiry at Excellence standard.

P - Problem

1. Rewrite the investigative question in your own words.
2. Identify the variables and their types.
 - Explanatory variable:
 - Response variable:
3. State a sensible hypothesis before analysis.

P - Plan

Describe how you will investigate the relationship.

Include:

1. What graphs you will use.

2. What numerical summaries you will use.
3. How you will decide whether the relationship is weak, moderate, or strong.
4. How you will check if unusual values affect your conclusion.

D - Data

1. Where does the data come from?
2. What is your sample size after data cleaning?
3. List two data-quality or representativeness limitations.

A - Analysis

Graphs and evidence

1. Attach or redraw your main scatterplot of birth weight vs mother age.
2. Describe the scatterplot using all of these words:

- direction
- form
- strength
- unusual features

3. Record key numerical evidence.

- Correlation (r): _____
- Regression slope: _____
- R^2 : _____

4. Explain what the slope means in context.

5. Explain what R^2 means in context.

Diagnostics and influence

6. What did your residual or diagnostic plots suggest about model quality?

7. Did influential points change your overall conclusion? Explain.

C - Conclusion

Write a final conclusion that:

1. Answers the investigative question directly.
2. Uses evidence from your graphs and model.
3. Interprets findings in context.
4. Critically evaluates limitations.
5. Suggests improvements or next steps.

Excellence Checklist

Tick each item when complete.

- ☐ I used precise statistical language correctly.
- ☐ I used both graph and numerical evidence.
- ☐ I made a contextual conclusion, not just a technical one.
- ☐ I discussed limitations and avoided causal overclaims.
- ☐ I suggested realistic improvements to the enquiry.